Build & Evaluate Linear Regression: California Housing

Author: Chayan Das | Date: 9th June

1. Introduction

This project implements a Machine Learning workflow—data loading, EDA, preprocessing, training, and evaluation—to predict California housing prices using the California Housing Dataset. This report details the end-to-end development of a linear regression model, aimed at establishing a baseline performance for predicting median house values in California. By leveraging statistical analysis and systematic feature engineering, we uncover key economic drivers influencing real estate pricing trends.

2. Dataset & Preprocessing

- **Data:** 20,640 records, 9 columns. Target: `MedHouseVal`.
- **Features:** MedInc, HouseAge, AveRooms, Population, AveOccup, Latitude, Longitude. (Removed `AveBedrms` due to weak correlation).

3. Methodology: Train-Test Split

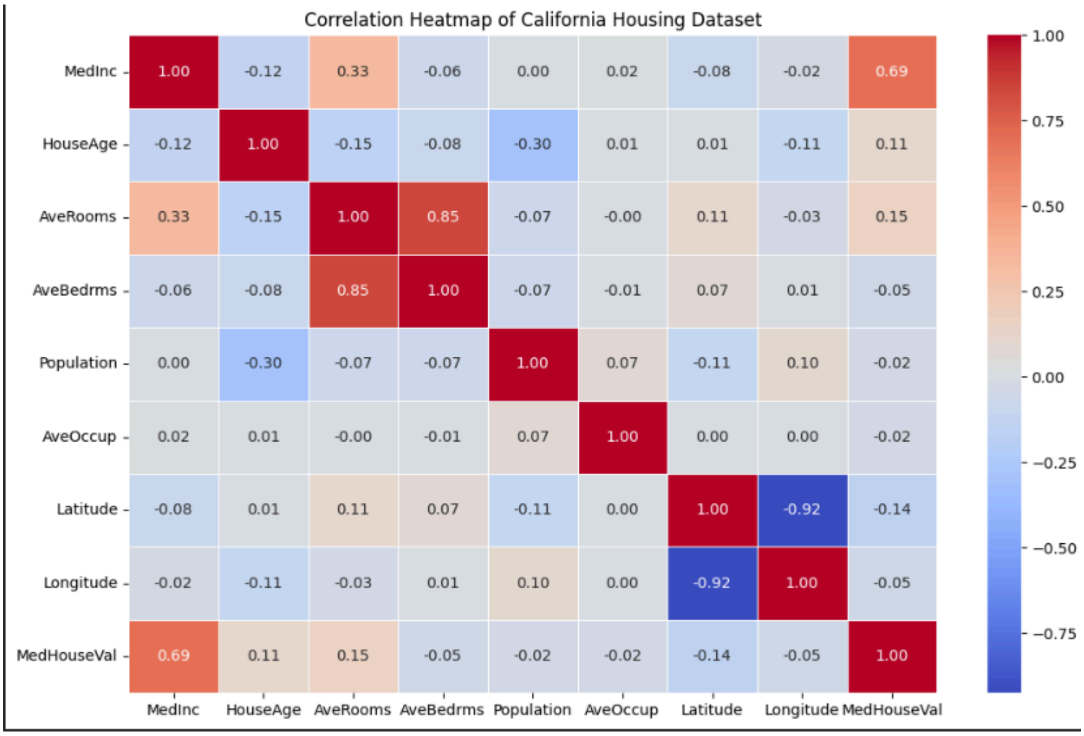
The dataset was partitioned using an 80/20 train-test split (random_state=42). This step is critical because splitting data into independent training and testing sets is essential to evaluate the model's ability to generalize to unseen data. This methodology helps in preventing overfitting and provides a reliable performance assessment by ensuring the model is tested on data it did not encounter during the training phase.

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

4. Exploratory Data Analysis (EDA)

- **Key Finding:** `MedInc` (Median Income) shows the strongest positive correlation (0.688) with house values.
- **Correlation Matrix:** Used to identify feature relationships and multicollinearity.

Feature	MedInc	AveRooms	HouseAge	Latitude	Longitude
Correlation	0.688	0.152	0.106	-0.144	-0.046



Note: These correlation values represent the Pearson correlation coefficient with the target variable MedHouseVal. Values near 1.0 indicate a strong positive relationship, values near -1.0 indicate a strong negative relationship, and values near 0 suggest no significant linear relationship.

5. Model Development & Evaluation

We utilized the Linear Regression model to capture the linear relationship between the housing features and median house values. The model uses the equation: Price = Intercept + (Coefficient_1 * Feature_1) + ... + (Coefficient_n * Feature_n). The calculated model parameters are:

- Intercept: -37.0345

- Coefficients: [4.3736e-01, 1.0521e-02, -1.0732e-01, 1.0000e-05, -3.8967e-03, -4.2131e-01, -4.3370e-01]

These coefficients quantify the impact of each feature on the predicted house value, while the intercept represents the baseline price when all other features are zero.

```
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Initialize and train the model
model = LinearRegression()
model.fit(X_train, y_train)

# Make predictions
y_pred = model.predict(X_test)

# Evaluate the model
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print("Coefficients of model are:", model.coef_)
print("Intercept of model is:", model.intercept_)

print(f"Mean Squared Error: {mse:.4f}")
print(f"Root Mean Squared Error: {rmse:.4f}")
print(f"R^2 Score: {r2:.4f}")
```

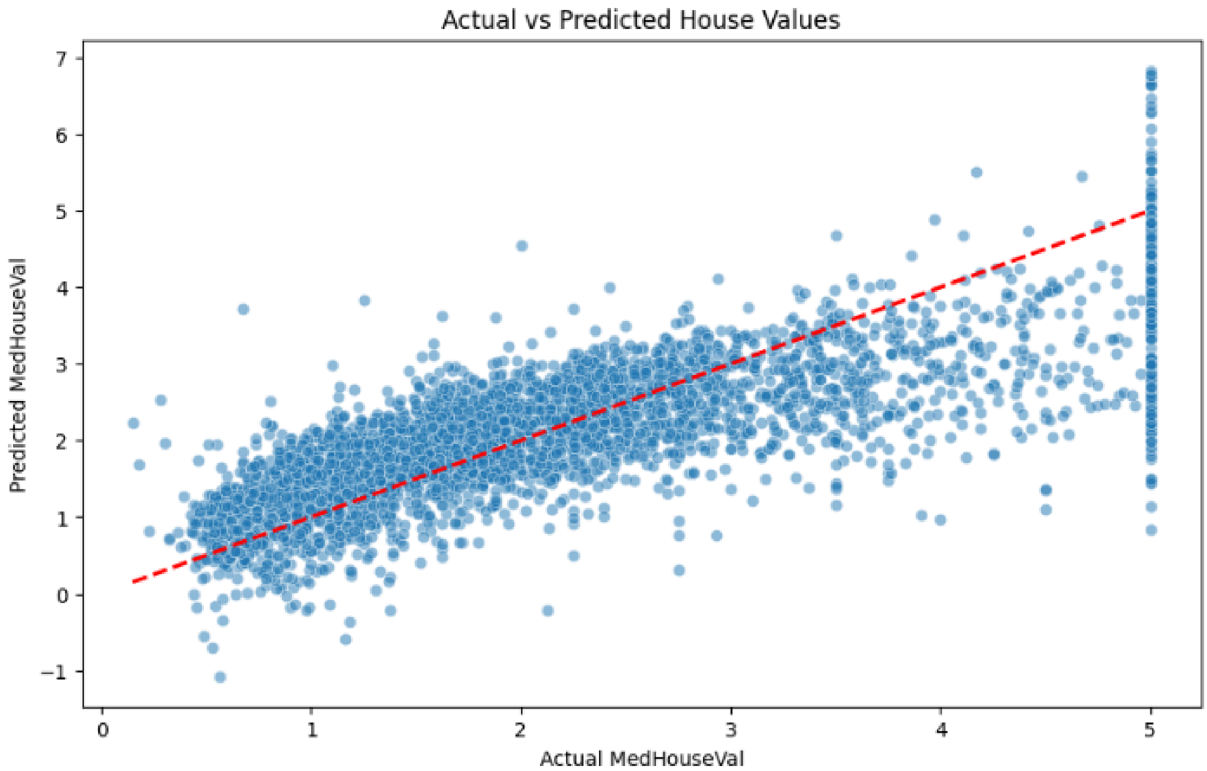
```
Coefficients of model are: [ 3.71939131e-01  9.79245076e-03  1.91164269e-02 -2.90695853e-06
 -3.23733409e-03 -4.57872368e-01 -4.64571653e-01]
Intercept of model is: -38.9758615951395
Mean Squared Error: 0.5473
Root Mean Squared Error: 0.7398
R^2 Score: 0.5823
```

Performance Metrics

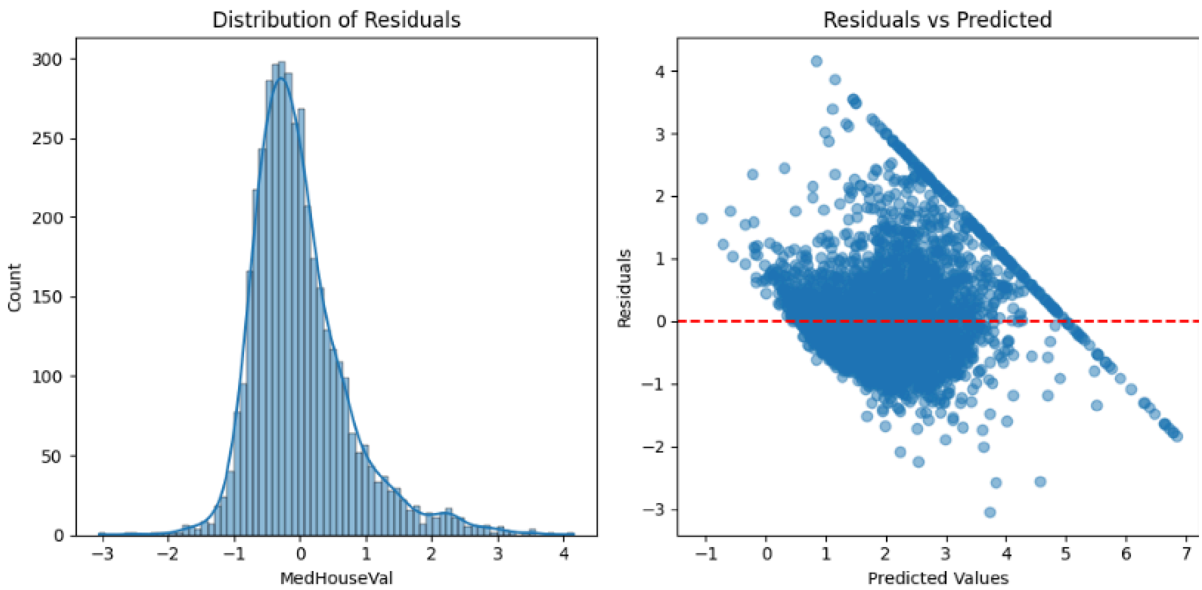
Metric	Training	Testing
RMSE	0.7317	0.7398
R^2 Score	0.5995	0.5823
MSE	-	0.5473

Visualization Analysis

- **Actual vs. Predicted:** Points align with the trend line, indicating moderate performance.



- **Residuals:** Centered around zero with no clear patterns, satisfying linear model assumptions.



6. Conclusion & Future Work

6.1 Conclusion

In this project, we successfully developed a baseline machine learning model to predict median housing values in California. The analysis identified Median

Income (MedInc) as the most critical determinant of housing prices, reinforcing established economic theory regarding real estate valuation. Our Linear Regression model achieved an R^2 score of 0.5823, indicating that approximately 58% of the variance in house prices is captured by the current feature set. The comparison between training and testing metrics confirms that the model is robust, demonstrating consistent performance without signs of overfitting. While Linear Regression provides a highly interpretable and efficient baseline for price estimation, the current residual patterns suggest that there is still significant room for capturing non-linear relationships and geographic complexities within the data.

6.2 Future Work

To enhance the predictive accuracy and generalizability of this model, the following improvements are proposed for future iterations:

- **Implementation of Non-Linear Algorithms:** Transitioning to advanced ensemble learning techniques, such as Random Forest, Gradient Boosting, or XGBoost, will likely allow the model to capture complex, non-linear patterns that simple linear models often miss.
- **Feature Engineering & Scaling:** Applying techniques like standard scaling or normalization could improve model convergence. Furthermore, creating interaction terms or polynomial features might better represent the complex dependencies between housing age, occupancy, and location.
- **Spatial Data Analysis:** Given that location (latitude and longitude) is a significant factor, incorporating geocoding or spatial clustering could refine the model's sensitivity to regional market variations.
- **Outlier Treatment:** Further investigation into extreme data points (outliers) in the target variable might help stabilize predictions and reduce residual error, leading to a more reliable model for real-world deployment.

Project Notebook

Access the full analysis and code here: [California Housing Price Prediction Notebook]